Principal Component Analysis

- Widely used technique for
  - dimensionality reduction
  - data compression (lossy)
  - feature extraction
  - data visualisation
- Can be defined in 2 ways
  - *Orthogonal projection* of data onto lower dimensional *linear* space (*principal subspace*) such that *variance of projected data is maximised*.
  - Linear projection that minimises average projection cost.
- Also called Karhunen-Loeve transform.

Principal Component Analysis

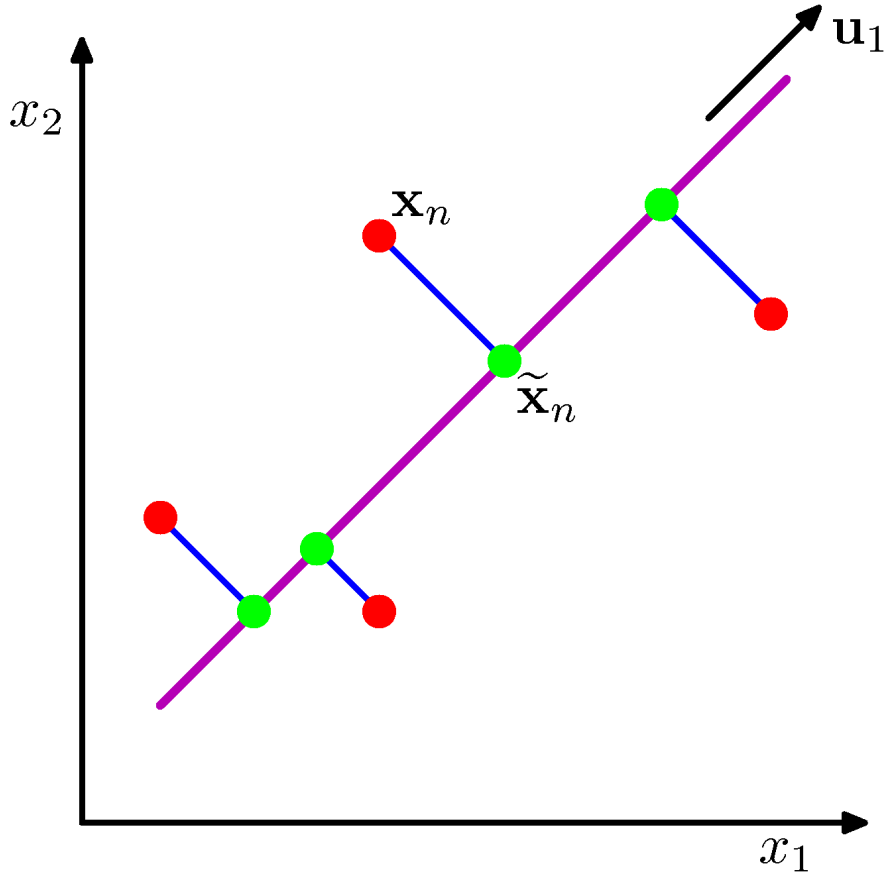


Figure: The two views of PCA. In this example for 2D data (in red), we want to find the direction vector $\mathbf{u}_1$ (in magenta) for which (**1**) the projections (in green) have maximum variance, or (**2**) the projection costs (lengths of blue lines) are minimum.

Maximum Variance Formulation of PCA

- Consider a set of signals $X = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$ where each $\mathbf{x}_i \in \mathbb{R}^D$.
- We have to find a vector $\mathbf{u} \in \mathbb{R}^D$ such that the variance of the projected data onto $\mathbf{u}$ is maximum.
- Projections of a data points $\mathbf{x}_i$ onto $\mathbf{u}$ are obtained via dot-products $\mathbf{u}^T\mathbf{x}_i$ for $i = 1, \ldots, N$.
- Mean of projected data is computed as $\mathbf{u}^T\bar{\mathbf{x}}$ where $\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$.
- Therefore, variance of projected data along direction $\mathbf{u}$ is computed as

$$\text{Var}(\mathbf{u}) = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^T\mathbf{x}_i - \mathbf{u}^T\bar{\mathbf{x}})^2$$

Maximum Variance Formulation of PCA

- Variance along $\mathbf{u}$ can be rewritten as the quadratic form

$$\begin{aligned}\text{Var}(\mathbf{u}) &= \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^T\mathbf{x}_i - \mathbf{u}^T\bar{\mathbf{x}})^2 = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^T\mathbf{x}_i - \mathbf{u}^T\bar{\mathbf{x}})(\mathbf{u}^T\mathbf{x}_i - \mathbf{u}^T\bar{\mathbf{x}})^T \\ &= \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^T\mathbf{x}_i - \mathbf{u}^T\bar{\mathbf{x}})(\mathbf{x}_i^T\mathbf{u} - \bar{\mathbf{x}}^T\mathbf{u}) = \frac{1}{N}\sum_{i=1}^{N}\mathbf{u}^T(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i^T - \bar{\mathbf{x}}^T)\mathbf{u} \\ &= \mathbf{u}^T \underbrace{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i^T - \bar{\mathbf{x}}^T)}_{S_{D\times D}}\mathbf{u} = \mathbf{u}^T S\mathbf{u}\end{aligned}$$

- We want to find the direction vector $\mathbf{u}$ that maximises the quadratic form $\mathbf{u}^T S\mathbf{u}$ where S is the *data covariance matrix*.
- Take-home Quiz 3: Prove that $\mathbf{u}^* = \arg\max_{\mathbf{u}} \mathbf{u}^T S\mathbf{u}$ is the eigenvector of S corresponding to the largest eigenvalue. (Hint: This is a constrained optimisation problem.)

Maximum Variance Formulation of PCA

- The eigenvector of S corresponding to the largest eigenvalue is called the *first principal component*.
- Additional principal components can be defined incrementally by *choosing each new projection direction as the one with maximum projected variance* among all directions *orthogonal to those already considered*.
- First M principal components correspond to the eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_M$ of S corresponding to the M largest eigenvalues $\lambda_1, \ldots, \lambda_M$. (Proof by induction in Exercise 12.1)
- Eigen-decomposition of $D \times D$ matrix has $O(D^3)$ complexity.
- For finding the first M eigenvectors only, there exist alternative methods such as the *power method* with $O(MD^2)$ complexity.

Choosing M

- Total variance of the data is given by the sum $V(D) = \sum_{i=1}^{D} \lambda_i$.
- By using the first M principal components, we capture variance amounting to $V(M) = \sum_{i=1}^{M} \lambda_i$.
- The remaining, uncaptured variance is called the *distortion measure* and is given by $J = \sum_{i=M+1}^{D} \lambda_i$.
- M can be chosen as the smallest integer for which $\frac{V(M)}{V(D)} > \tau$ where $0 < \tau \leq 1$.
- For example, $\tau = 0.95$ corresponds to retaining 95% of the total variance after projection.

Choosing M

- Even for $\tau = 1$, it is often observed that $M < D$.
- This shows that the *intrinsic dimensionality* of D-dimensional data is often less than D.
- Therefore, by working in this lower-dimensional space we do not loose any variations in the data.

PCA for high-dimensional data

- N points in $\mathbb{R}^D$ define an $N-1$ dimensional linear subspace.
- If $N < D$, the $D \times D$ covariance matrix S will have rank (= number of non-zero eigenvalues) at most $N-1$.
- The remaining $D-(N-1)$ eigenvalues of S will all be 0.
- So we should not compute more than $N-1$ eigenvectors.
- Projecting onto $M > N-1$ eigenvectors *does not imply* dimensionality reduction.
- The $N < D$ scenario occurs often. For example, in a dataset of $N = 100000$ RGB images of size 640×480, $D = 640 * 480 * 3 = 921600 >> N$.
- The $O(D^3)$ scaling also makes computing the eigenvectors of S impractical for large D.

PCA for high-dimensional data

- So we use a clever trick.
- Let $\tilde{\mathbf{X}}$ be the *data centered design matrix*.

$$\tilde{\mathbf{X}} = \begin{bmatrix} (\mathbf{x}_1 - \bar{\mathbf{x}})^T \\ (\mathbf{x}_2 - \bar{\mathbf{x}})^T \\ \vdots \\ (\mathbf{x}_N - \bar{\mathbf{x}})^T \end{bmatrix}$$

- We can write the data covariance matrix as

$$S = \sum_{i=1}^{N} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i^T - \bar{\mathbf{x}}^T) = \frac{1}{N} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$$

PCA for high-dimensional data

- The eigenvector equation can be written as

$$\begin{aligned} S\mathbf{u}_i = \lambda_i \mathbf{u}_i &\implies \frac{1}{N}\tilde{\mathbf{X}}^T\tilde{\mathbf{X}}\mathbf{u}_i = \lambda_i \mathbf{u}_i \\ &\implies \frac{1}{N}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^T\tilde{\mathbf{X}}\mathbf{u}_i = \lambda_i \tilde{\mathbf{X}}\mathbf{u}_i \\ &\implies \frac{1}{N}\tilde{\mathbf{X}}\tilde{\mathbf{X}}^T\mathbf{v}_i = \lambda_i \mathbf{v}_i \end{aligned} \tag{1}$$

  which shows that λ_i and $\mathbf{v}_i$ are eigenvalues and eigenvectors of the *smaller $N \times N$ matrix* $\tilde{\mathbf{X}}\tilde{\mathbf{X}}^T$.

- But notice that λ_i was also the eigenvalue of the original covariance matrix S. So we have found the eigenvalues of S in $O(N^3)$.

PCA for high-dimensional data

- To obtain the eigenvectors $\mathbf{u}_i$, pre-multiply both sides of Equation (1) by $\tilde{\mathbf{X}}^T$ to obtain

$$\left(\frac{1}{N}\tilde{\mathbf{X}}^T\tilde{\mathbf{X}}\right)\left(\tilde{\mathbf{X}}^T\mathbf{v}_i\right) = \lambda_i\left(\tilde{\mathbf{X}}^T\mathbf{v}_i\right)$$

  which shows that $\tilde{\mathbf{X}}^T\mathbf{v}_i$ is an eigenvector of S with eigenvalue λ_i.
- So the original eigenvectors are obtained as

$$\mathbf{u}_i = \frac{\tilde{\mathbf{X}}^T\mathbf{v}_i}{||\tilde{\mathbf{X}}^T\mathbf{v}_i||} = \frac{\tilde{\mathbf{X}}^T\mathbf{v}_i}{\sqrt{N\lambda_i}}$$

  Show that $||\tilde{\mathbf{X}}^T\mathbf{v}_i|| = \sqrt{N\lambda_i}$.
- So the eigen-decomposition of the $D \times D$ covariance matrix S can be achieved in $O(N^3)$.

PCA for high-dimensional data

Summary

- When $N < D$, construct the $N \times N$ matrix $\tilde{\mathbf{X}}\tilde{\mathbf{X}}^T$ and compute its eigenvalues λ_i and eigenvectors $\mathbf{v}_i$.
- Eigenvalues of S are also λ_i.
- Eigenvectors of S are obtained as

$$\mathbf{u}_i = \frac{\tilde{\mathbf{X}}^T \mathbf{v}_i}{\sqrt{N\lambda_i}}$$

- We now look at some applications of PCA.
- These include
  - Compression
  - Pre-processing of data
  - Visualization of data
  - Classification

Compression

- When data point $\mathbf{x}$ is projected onto the i-th principal component, coefficient of projection is given by

$$\alpha_i = (\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{u}_i$$

- Consider projections $\alpha_1, \ldots, \alpha_M$ onto the first M principal components where $M < D$.
- Reconstruction $\hat{\mathbf{x}}$ from these M scalar coefficients can be obtained as

$$\hat{\mathbf{x}} = \bar{\mathbf{x}} + \sum_{i=1}^{M} \alpha_i \mathbf{u}_i$$

Compression

- This dimensionality reduction represents *compression* from $\mathbb{R}^D$ to $\mathbb{R}^M$.
- In $\mathbb{R}^D$, N data points require storing ND values.
- After compression using the first M principal components, the N data points require storing $NM + D$ values. (Why $+D$?)
- You will implement compression via PCA in Project 4a when you regenerate Bishop's Figure 12.5.

Data pre-processing

- Sometimes different dimensions of data have different units or significantly different variability.
  - $\mathbf{x} = [\text{time (seconds), speed (mph), fuel consumption (liters)}]^T$.
  - $\mathbf{x} = [\text{time between earthquakes, duration of earthquake}]^T$.
- Averaged over the whole dataset, every component of $\mathbf{x}$ will have a different mean and different variance.
- Effectiveness of subsequent algorithms can be diminished due to such variability.
- Non-PCA solution: *Standardize* the data using $y_{ni} = \frac{x_{ni} - \bar{x}_i}{\sigma_i}$.
- Individual components of the transformed data $\mathbf{y}_1, \ldots, \mathbf{y}_N$ will now have *zero-mean and unit-variance*.
- However different components y_{ni} and y_{nj} can still be correlated.

Data pre-processing
Whitening

- A better PCA-based solution, known as *whitening* or *sphereing* transforms the data as

$$\mathbf{y}_n = \mathbf{L}^{-\frac{1}{2}}\mathbf{U}^T(\mathbf{x}_n - \bar{\mathbf{x}})$$

  where $\mathbf{L}$ is a $D \times D$ diagonal matrix of D eigenvalues λ_i of S and $\mathbf{U}$ is an orthogonal $D \times D$ matrix with columns given by the corresponding eigenvectors $\mathbf{u}_i$.
- Easy to show that transformed data $\mathbf{y}_1, \ldots, \mathbf{y}_N$ has zero-mean and its covariance matrix $\frac{1}{N}\sum_{n=1}^{N}\mathbf{y}_n\mathbf{y}_n^T$ equals $\mathbf{I}_{D\times D}$. **Show it.**
- So, individual components of the transformed data $\mathbf{y}_1, \ldots, \mathbf{y}_N$ will now have *zero-mean and unit-covariance*.

Visualization

- Project data onto the first 1, 2, or 3 principal components and visualise these projected coefficients.

Classification via PCA

- Training
  1. Compute eigen-decomposition of the complete training data $\mathbf{x}_1, \ldots, \mathbf{x}_N$.
  2. Form orthogonal eigen-basis from the first M principal components.
  3. Project each mean-subtracted training sample $\mathbf{x}_n - \bar{\mathbf{x}}$ onto the eigen-bases to obtain projected coefficients $\phi_n \in \mathbb{R}^M$.
- Testing
  1. Project mean-subtracted test sample $\mathbf{x} - \bar{\mathbf{x}}$ onto the eigen-bases to obtain projected coefficients $\phi \in \mathbb{R}^M$.
  2. Compute Euclidean distance of coefficients ϕ from each of the coefficients ϕ_n of the training samples.
  3. Class of $\mathbf{x}$ is the class of the nearest neighbour nn from the training samples where

$$nn = \arg\min_n ||\phi - \phi_n||^2$$

- This is essentially nearest neighbour classification in $\mathbb{R}^M$ instead of $\mathbb{R}^D$.